Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/07/2023
PI Name:	Foster, Jamie Ph.D.		
Project Title:	Effects of Modeled Microgravity on the Induction of Bacteria-Induced Apoptosis During Animal Development		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	EDITOR'S NOTE: Congressional district is for University of Florida, Gainesville. Address is PI's physical address.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/26/2018	**End Date:**	07/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2023 per NSSC information (Ed., 7/27/22) NOTE: End date changed to 7/31/2022 per NSSC information (Ed., 9/15/21)		
Key Personnel Changes/Previous PI:	Madeline Vroom has now received her Ph.D. and graduated in Fall of 2021. A new graduate student Joseph Emhof has now joined my lab and will pick up on the final analyses of this work and will continue to use this model system the results from Objectives 2 and 3 for his dissertation work. Additionally, a post-doctoral fellow used some left over animals that were not used in the proposed experiments and conducted some additional stretch goals as outline in the final report.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K1465		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Long-duration space flight imposes numerous physiological challenges to the human body, such as bone loss, dysregulation of the immune system, and radiation exposure. Although considerable progress has been made in delineating these biological phenomena, one area of research that has received little attention is the impact that space flight has on those commensal, non-pathogenic microbes that associate with human tissues. To address this gap in knowledge we propose to examine how microgravity alters normal bacteria-induced development in animal tissues. Specifically, we propose to elucidate the pathways associated with bacteria-induced apoptosis, a type of programmed cell death that is critical in maintaining tissue homeostasis and the immune response of animals. To address this goal, the monospecific symbiosis between the bobtail squid Euprymna scolopes and its luminescent partner Vibrio fischeri will be used as a model system to examine the effects of modeled microgravity (i.e., High Aspect Ratio Vessel) on bacteria-induced apoptosis in the host animal. The central hypothesis of this project is that microgravity alters the extrinsic and intrinsic signaling pathways associated with bacteria-induced apoptosis in the host animal. Our previous studies have shown that modeled microgravity causes an acceleration of bacteria-induced apoptosis in the host; however, the mechanisms underlying the developmental acceleration is unknown. To address this issue we propose the following specific objectives: Objective 1. Delineate the impact of modeled microgravity on the shedding of apoptosis-inducing lipopolysaccharide by the bacterial symbiont. We hypothesize that the acceleration of cell death in the host tissues is due to the increased shedding of bacterial lipopolysaccharide (LPS), thereby triggering an upregulation of genes associated with the extrinsic apoptosis pathway. To address this objective we will use a three-pronged approach to examine the controls on LPS-shedding and identify the host molecular response to the shedding using NanoString, western blotting, and immunocytochemistry under modeled microgravity conditions. Objective 2. Elucidate changes in activation and mediation of the extrinsic and intrinsic apoptosis pathways associated with bacteria-induced apoptosis in microgravity. By data mining currently available transcriptomes, we will monitor the differential expression of potential target ligands, receptors, adaptor proteins, and inhibitors of apoptosis. Objective 3. Examine the onset of caspase activity during accelerated apoptosis in host light organ under simulated microgravity conditions and attempt to modulate its onset. We will examine the differential expression of caspases within the light organ under modeled microgravity conditions and anticipate that the caspases are highly integrated with the innate immune response of the host. Additionally, we will attempt to modulate the protease activity using a range of inhibitors to potentially mitigate the microgravity-induced effects on bacteria-induced apoptosis in the host tissues. Significance of Project - The proposed work has several implications for the field of space biology and directly relates to NASA Space Biology Science Plan and Decadal Survey, specifically the Microbiology (MB) and Cell Molecular Biology (CMB) program elements. First, by examining how the normal bacteria-induced development of animals is mediated under microgravity conditions, we can begin to determine the sensitivity of these associations to perturbations in the space environment. Secondly, by removing gravity as a constant we can address the question of whether gravity obscures aspects of bacteria-induced animal development that may otherwise go undetected under terrestrial conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed work has several implications for the field of space biology and life on Earth. First, by examining how healthy animal-microbe interactions are initiated and maintained under microgravity conditions, we were able to begin to determine the sensitivity of these associations to perturbations in the space environment. There is a gap in our understanding of the effects that the space environment has on the microbiome of animals. As there are more than 2000 identified bacterial species that form associations with humans, there are fewer than 100 species identified as human pathogens, yet most microgravity studies have focused on pathogens. This leaves a large array of undescribed commensal organisms that form interactions with animal cells, which may, or may not, be impacted by microgravity conditions. Here, in this project, we asked the question of how does microgravity alter key bacteria-induced developmental events within the host animal? In other words, what happens to mutualistic associations when gravity is removed? Secondly, by removing gravity as a constant we were able to the question of whether gravity obscures aspects or cues of bacteria-induced animal development that may otherwise go undetected under terrestrial conditions. These new insights from simplified animal model systems can be applied to our understanding of human physiology in the space environment.
	Major takeaways from the completed work: • In Vibrio fischeri (V. fischeri), modeled microgravity causes the increased shedding of the endotoxin lipopolysaccharide via the increased production of outer membrane vesicles (OMVs). • The cell membrane of the bacterial symbiont Vibrio fischeri is negatively impacted by modeled microgravity making it more susceptible to cell lysis. • Perturbations, such as the stress of modeled microgravity, amplifies the expression of apoptosis genes in the host animal Euprymna scolopes, thereby accelerating normal bacteria-induced developmental pathways in the host. • Model microgravity increases the levels of apoptosis in the host animals through caspase-mediated pathways. • The increases in apoptosis levels under modeled microgravity can be mitigated pharmacologically using caspase inhibitors. • Modeled microgravity can alter the colonization behaviors of symbiotic bacteria on host epithelial cells and their population structure within the host animal tissues. Specific findings from the completed work: Objective 1. Delineate the impact of modeled microgravity on the shedding of apoptosis-inducing lipopolysaccharide (LPS) by the bacterial symbiont. 1. Low-shear modeled microgravity (LSMMG) increases the release of apoptosis-inducing LPS by V. fischeri. Using two different analysis methods (i.e., Limulus Amebocyte Lysate Endotoxin Assay Kit, and Purpald Assay), the results indicated that there was a significant increase in the amount of shed LPS from cells grown under modeled microgravity conditions. 2. The increase shedding of LPS is correlated with flagellar rotation. Using two V. fischeri mutants that have either no flagella or that are defective in rotating their flagella, we showed that the lack of motility by V. fischeri significantly

Task Progress:

decreased the amount of LPS shedding under both gravity and modeled microgravity conditions. However, even though the amount of LPS was significantly lower, there was a high amount of shedding in modeled microgravity exposed bacteria, suggesting that flagellar rotation is not the only factor associated with the increased LPS shedding in the simulated microgravity conditions.

3. There was increased production of outer membrane vesicles, and the size of the OMVs was larger under modeled microgravity conditions. Cells grown under LSMMG conditions were harvested and examined for OMVs and their sizes. We hypothesized that the LPS being shed by V. fischeri is located on the OMVs and were able to isolate OMVs and demonstrated, using NanoSight and LPS labeling, the increase of OMV size under modeled microgravity.

4. The membranes of the V. fischeri cells are weaker and more sensitive to the antibiotic polymyxin B under modeled microgravity conditions. Results of our analysis suggest under modeled microgravity the membranes of the bacteria are compromised.

Implications: The implications of this work suggest that microgravity may be causing the increased shedding of bacterial endotoxin and, depending on the bacterium, this could be problematic for host organisms.

Objective 2. Elucidate changes in activation and mediation of the extrinsic and intrinsic apoptosis pathways associated with bacteria-induced apoptosis in microgravity.

1. Perturbations, such as the stress of low sheer modeled microgravity, can amplify the apoptosis cascade in host animal tissues. Using previously generated RNAseq and genome data, we generated the first map of putative apoptosis pathways in E. scolopes. Additionally, using the gene expression approach called NanoString, we targeted 42 specific apoptosis genes to monitor the differential expression of the apoptosis genes during the stress of modeled microgravity. Our results demonstrated a very distinctive pattern of expression of apoptosis genes in response to the stress of microgravity.

2. Results suggest that the increased shedding of LPS under modeled microgravity is also coupled with an increase in the receptors for LPS. Analysis of the genes that encode for LPS-binding protein (lbp) receptors using the NanoString technique revealed an increase in the abundance of the expression of lbp1 and lpb3 under modeled microgravity at the peak of apoptosis seen in the host animals. The abundance of LPS shedding and LPS-binding receptors may correspond to the increase in apoptotic cell death observed during modeled microgravity conditions.

3. Results indicate that modeled microgravity causes increase in both the extrinsic and intrinsic apoptosis pathways. One of the goals of the project was to determine if the increase in apoptosis observed under modeled microgravity was coming from extrinsic (e.g., bacterial signals) or intrinsic stressors (e.g., mitochondrial, or oxidative stress). Results from the NanoString gene expression analysis revealed that modeled microgravity caused an increase in the expression of both types of apoptosis gene pathways.

Implications: Modeled microgravity is causing an increase in bacteria-induced apoptosis, thereby accelerating normal developmental pathways in the host animal.

Objective 3. Examine the onset of caspase activity during accelerated apoptosis in host light organ under simulated microgravity conditions and attempt to modulate its onset.

1. Caspase gene expression increased significantly in modeled microgravity compared to gravity controls. Through the research associated with Objective 2, we found several candidate genes for caspase enzymes, which are essential components of the apoptosis. As part of Objective 3, we examined the activity of these enzymes in the presence and absence of the symbiont under modeled microgravity and unit gravity conditions. This expression occurred significantly earlier than normal expression under gravity conditions.

2. Structure analysis of the protein architecture of the squid caspases found several unique isoforms of the caspase proteins compared to other animals and cephalopods. Most of the initiator caspases and all the observed executioner caspases shared homology with other animals, suggesting these enzymes are highly conserved in animals.

3. Caspase-mediated apoptosis caused by the modeled microgravity could be mitigated by the addition of various caspase inhibitors. Using a series of pharmacological inhibitors, we could mitigate some, but not all, of the induced apoptotic cell death. The results suggest that additional apoptosis pathways may be impacted by modeled microgravity

Implications: Modeled microgravity appears to increase levels of apoptosis in the host tissues through a caspase-mediated pathway. The microgravity-induced increase in apoptosis levels can, however, be mitigated pharmacologically using various types of inhibitors to caspases.

Additional research findings not part of the original three objectives derived from the project:

Since we had additional animals left over for each animal collection trip, we were able to do additional experiments because of this grant that were not part of the original goals. As a result, we were able to explore whether modeled microgravity alters the normal colonization behaviors of the symbiont when colonizing the host epithelial tissues of the host squid. We found the following results:

1. Modeled microgravity did not alter the strain behavior of the symbiont Vibrio fischeri. Often, different strains of the same species have different colonization behaviors that can shape the population structure within a host organism. In V. fischeri there are dominant (i.e., occur as a single population) and sharing strains (i.e., different strains can mix) that colonize the host animal epithelial cells. Exposing the strains to modeled microgravity and allowing them to colonize the host did not change their fundamental colonization behaviors.

2. Sharing strains showed decreased population levels under modeled microgravity. When squid were colonized by sharing strains of V. fischeri, the number of CFU per light organ was significantly decreased in modeled microgravity compared to unit gravity. This phenotype was not observed when the squids were colonized with a dominant strain.

3. Bacterial aggregation was not significantly impacted under modeled microgravity stress. Since bacterial aggregation has been linked to the efficiency of host light organ colonization by multiple studies, we assessed whether mutants with altered aggregation phenotypes would be impacted by the stress of modeled microgravity; however, no significant differences were observed between the number of bacteria in squid that were singly colonized for either aggregation mutant under LSMMG or gravity conditions. The results suggest that the stress of modeled microgravity doesn’t alter the colonization of the light organ by these aggregation mutant strains.

	Implications: To our knowledge, this was the first description of the initial assembly and structure of a microbial symbiotic population in an animal host under modeled microgravity. Our data suggest that environmental stresses, such as modeled microgravity, can impact the symbiont density within host tissues in a strain-specific manner, potentially impacting the health of the host organism.
Bibliography Type:	Description: (Last Updated: 04/20/2026)
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